

CHAPTER : 2

2.0 Introduction

The Entity Analysis for the **Sephora Beauty E-Commerce Website** identifies the major entities required to design an efficient and well-structured database for the online shopping platform. These entities represent the core components of the system, including customer management, product catalog, categories, brands, shopping cart, wishlist, order processing, payment management, shipment tracking, and administrator management.

Each entity contains a set of attributes that describe the information stored in the database. Every entity is assigned a Primary Key (PK) to uniquely identify each record, while Foreign Keys (FK) are used to establish relationships between different entities. These relationships ensure proper connectivity among various modules of the application and maintain data integrity.

The Entity Analysis helps reduce data redundancy, improves consistency, supports efficient data retrieval, and ensures secure transaction processing. It serves as the foundation for designing the relational database that powers the Sephora Beauty E-Commerce Website and enables reliable management of products, customers, orders, and payments.

2.1 Customer

The **Customer** entity stores the information of users who register on the Sephora Beauty E-Commerce Website. Customers can create accounts, log in securely, browse beauty products, search and filter products, add products to the wishlist or shopping cart, place orders, make online payments, and track their deliveries. Each customer is assigned a unique Customer_ID to identify individual records and maintain purchase history.

Attribute	Data Type	Key	Description
Customer_ID	INT	PK	Unique customer ID
First_Name	VARCHAR(50)	-	Customer first name
Last_Name	VARCHAR(50)	-	Customer last name
Gender	VARCHAR(10)	-	Gender
Email	VARCHAR(100)	-	Email address
Phone	VARCHAR(15)	-	Contact number
Password	VARCHAR(255)	-	Encrypted login password

Attribute	Data Type	Key	Description
Address	TEXT	-	Customer address
City	VARCHAR(50)	-	Customer city
State	VARCHAR(50)	-	Customer state
Pincode	VARCHAR(10)	-	Postal code
Registration_Date	DATE	-	Account creation date

2.2 Admin

The **Admin** entity represents the administrator responsible for managing the entire Sephora Beauty E-Commerce Website. The administrator manages products, brands, categories, customer accounts, inventory, orders, payments, and shipment details. This entity ensures efficient control over the website's operations and provides secure access to administrative functions.

Attribute	Data Type	Key	Description
Admin_ID	INT	PK	Unique administrator ID
Admin_Name	VARCHAR(100)	-	Administrator name
Email	VARCHAR(100)	-	Email address
Phone	VARCHAR(15)	-	Contact number
Username	VARCHAR(50)	-	Login username
Password	VARCHAR(255)	-	Encrypted login password
Role	VARCHAR(30)	-	Administrator role

2.3 Category

The **Category** entity stores the various categories of beauty and cosmetic products available on the website. Categorizing products enables customers to browse products easily and improves product organization within the system.

Attribute	Data Type	Key	Description
Category_ID	INT	PK	Unique category ID
Category_Name	VARCHAR(100)	-	Category name
Description	TEXT	-	Category description
Status	VARCHAR(20)	-	Active or inactive status

2.4 Brand

The **Brand** entity contains details about cosmetic brands available on the Sephora Beauty E-Commerce Website. Customers can filter and search products based on their preferred brands, making it easier to find suitable beauty products.

Attribute	Data Type	Key	Description
Brand_ID	INT	PK	Unique brand ID
Brand_Name	VARCHAR(100)	-	Brand name
Country	VARCHAR(50)	-	Country of origin
Description	TEXT	-	Brand description
Status	VARCHAR(20)	-	Active or inactive status

2.5 Product

The **Product** entity stores detailed information about all beauty and cosmetic products available on the website. Every product belongs to a category and a brand. Customers can view product information, compare products, check prices, read descriptions, and purchase products through the online shopping platform.

Attribute	Data Type	Key	Description
Product_ID	INT	PK	Unique product ID
Product_Name	VARCHAR(150)	-	Product name
Category_ID	INT	FK	References Category

Attribute	Data Type	Key	Description
Brand_ID	INT	FK	References Brand
Price	DECIMAL(10,2)	-	Product price
Stock	INT	-	Available stock quantity
Description	TEXT	-	Product description
Ingredients	TEXT	-	Product ingredients
Skin_Type	VARCHAR(50)	-	Suitable skin type
Image_URL	VARCHAR(255)	-	Product image
Rating	DECIMAL(2,1)	-	Customer rating
Status	VARCHAR(20)	-	Product availability

2.12 Relationships

Entity 1	Relationship	Entity 2	Cardinality
Customer	Places	Orders	1 : M
Customer	Adds	Shopping Cart	1 : M
Customer	Saves	Wishlist	1 : M
Category	Contains	Product	1 : M
Brand	Belongs To	Product	1 : M
Product	Added To	Shopping Cart	1 : M
Product	Added To	Wishlist	1 : M
Orders	Contains	Order Details	1 : M
Product	Appears In	Order Details	1 : M

Entity 1	Relationship	Entity 2	Cardinality
Orders	Has	Payment	1 : 1
Orders	Has	Shipment	1 : 1
Admin	Manages	Products	1 : M
Admin	Manages	Categories	1 : M
Admin	Manages	Brands	1 : M
Admin	Manages	Orders	1 : M

2.13 Conclusion

The Entity Analysis of the Sephora Beauty E-Commerce Website identifies the essential entities, their attributes, keys, and relationships required for developing an efficient and reliable database system. The relationships established among customers, products, categories, brands, shopping carts, wishlists, orders, payments, shipments, and administrators ensure proper data organization and integrity. This analysis minimizes data redundancy, supports efficient data management, and provides a strong foundation for designing the relational database that powers the e-commerce platform. It also enhances system performance, maintainability, and scalability, ensuring a secure and user-friendly online shopping experience.